Class11

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Background

We saw last day that the main repository for biomolecular structure (the PDB database) only has ~250,000 entries.

UniProtKB (the main protein sequence database) has over 200 million re

AlphaFold

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

This major breakthrough (Figure 1) promises to place Molecular Biology in a new era where we can visualize, analyze and interpret the structures and functions of all proteins.

The EBI AlphaFold database

The EBI alphafold database contains lost of computed structure models. It is increasingly likely that the structure you are interested in is already in this database. < https://alphafold.ebi.ac.uk >

There are 3 major outputs from AlphaFold

1. A model of structure in **PDB** format.
2. a **pLDDT score**: that tells us how confident the model is for a given residue in your protein (High values are good, above 70)
3. a **PAE score** that tells us about protein packing quality.

If you can't find a matching entry for the sequence you are interested in AFDB you can run AlphaFold yourself…

Running AlphaFold

We will use < http://github.com/sokrypton/ColabFold >

Figure from AlphaFold here!

Interpreting Results

Custom analysis of resulting models

We can read all the AlphaFold results into R and do more quantitative analysis than just viewing the structures in Mol-star:

Read all the PDB models:

```
results_dir <- "hivpr_23119/"
```

```
# File names for all PDB models
pdb_files <- list.files(path=results_dir,
pattern="*.pdb",
full.names = TRUE)
# Print our PDB file names
basename(pdb_files)
```

```
[1] "hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3dview)
#view.pdbs(pdbs)
```

```
library(bio3d)
# Read all data from Models
# and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
```

```
hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
pdb/seq: 2   name: hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
pdb/seq: 3   name: hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
pdb/seq: 4   name: hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
pdb/seq: 5   name: hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3
```

```
pdbs
```

```
                              1        .         .         .         .         50
[Truncated_Name:1]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                              **************************************************
                              1        .         .         .         .         50

                             51        .         .         .         .         100
[Truncated_Name:1]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                              **************************************************
                             51        .         .         .         .         100

                            101        .         .         .         .         150
[Truncated_Name:1]hivpr_2311  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]hivpr_2311  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]hivpr_2311  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]hivpr_2311  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]hivpr_2311  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                              **************************************************
                            101        .         .         .         .         150
```

```
                              151        .         .         .         .       198
[Truncated_Name:1]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                               **********************************************
                              151        .         .         .         .       198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

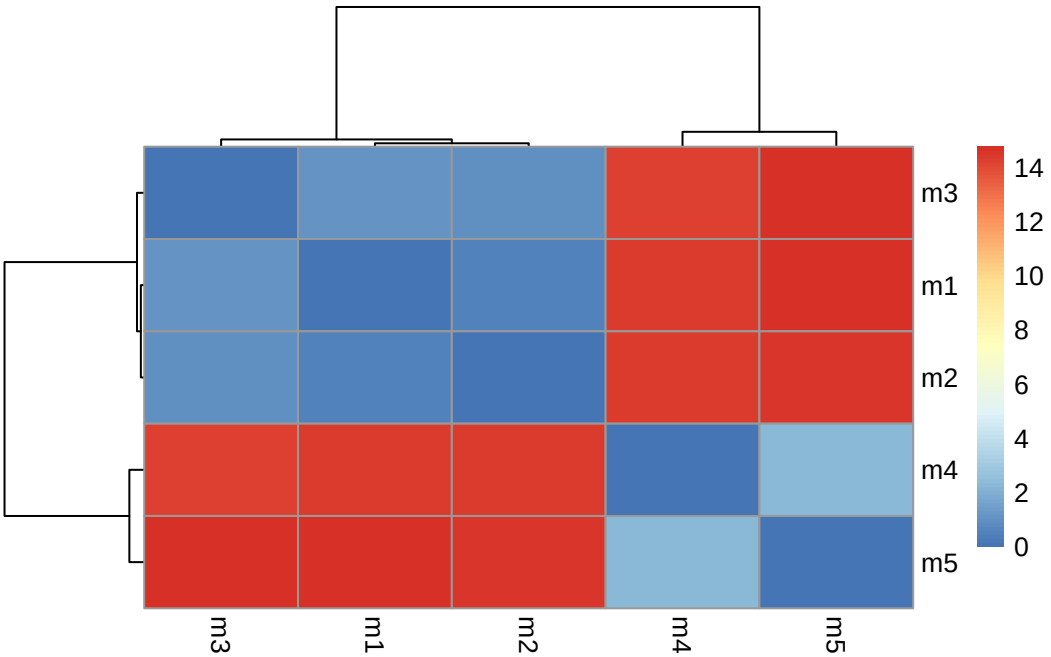
```r
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.754
```

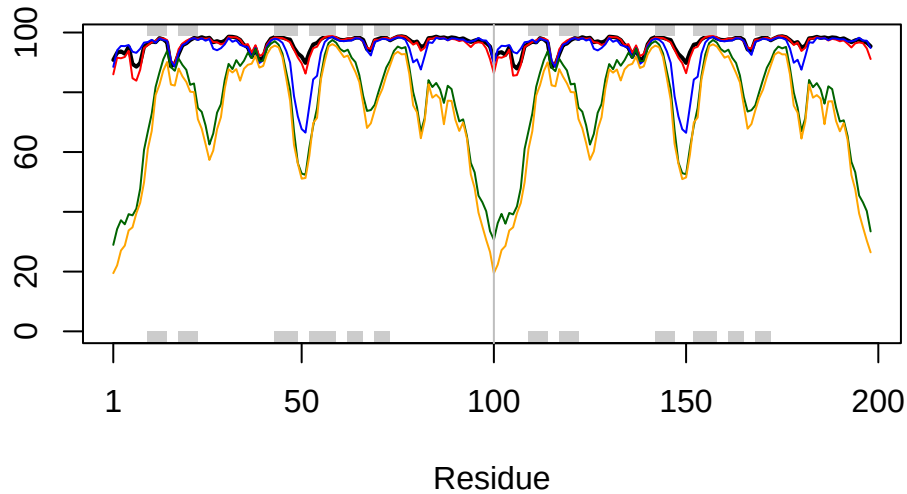
```r
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```




```
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```




```r
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 9836.196
core size 196 of 198  vol = 6800.793
core size 195 of 198  vol = 1322.091
core size 194 of 198  vol = 1032.249
core size 193 of 198  vol = 944.133
core size 192 of 198  vol = 894.048
core size 191 of 198  vol = 829.724
core size 190 of 198  vol = 766.628
core size 189 of 198  vol = 728.681
core size 188 of 198  vol = 693.317
core size 187 of 198  vol = 656.166
core size 186 of 198  vol = 622.063
core size 185 of 198  vol = 586.601
core size 184 of 198  vol = 565.358
core size 183 of 198  vol = 532.747
core size 182 of 198  vol = 511.421
core size 181 of 198  vol = 489.077
core size 180 of 198  vol = 468.599
core size 179 of 198  vol = 449.222
core size 178 of 198  vol = 433.122
```

```
core size 177 of 198  vol = 418.651
core size 176 of 198  vol = 405.069
core size 175 of 198  vol = 391.855
core size 174 of 198  vol = 381.006
core size 173 of 198  vol = 371.54
core size 172 of 198  vol = 355.539
core size 171 of 198  vol = 345.156
core size 170 of 198  vol = 335.982
core size 169 of 198  vol = 325.275
core size 168 of 198  vol = 313.603
core size 167 of 198  vol = 302.908
core size 166 of 198  vol = 293.343
core size 165 of 198  vol = 284.547
core size 164 of 198  vol = 277.932
core size 163 of 198  vol = 269.592
core size 162 of 198  vol = 258.167
core size 161 of 198  vol = 246.856
core size 160 of 198  vol = 239.073
core size 159 of 198  vol = 234.174
core size 158 of 198  vol = 226.309
core size 157 of 198  vol = 221.212
core size 156 of 198  vol = 214.951
core size 155 of 198  vol = 206.027
core size 154 of 198  vol = 197.301
core size 153 of 198  vol = 190.889
core size 152 of 198  vol = 185.403
core size 151 of 198  vol = 179.154
core size 150 of 198  vol = 174.709
core size 149 of 198  vol = 167.269
core size 148 of 198  vol = 160.866
core size 147 of 198  vol = 156.173
core size 146 of 198  vol = 148.395
core size 145 of 198  vol = 143.389
core size 144 of 198  vol = 138.428
core size 143 of 198  vol = 132.99
core size 142 of 198  vol = 126.924
core size 141 of 198  vol = 121.257
core size 140 of 198  vol = 116.407
core size 139 of 198  vol = 112.241
core size 138 of 198  vol = 107.813
core size 137 of 198  vol = 104.807
core size 136 of 198  vol = 100.968
core size 135 of 198  vol = 97.21
```

```
core size 134 of 198  vol = 92.54
core size 133 of 198  vol = 87.964
core size 132 of 198  vol = 85.621
core size 131 of 198  vol = 81.514
core size 130 of 198  vol = 77.627
core size 129 of 198  vol = 74.881
core size 128 of 198  vol = 72.642
core size 127 of 198  vol = 70.352
core size 126 of 198  vol = 68.627
core size 125 of 198  vol = 66.349
core size 124 of 198  vol = 64.019
core size 123 of 198  vol = 61.763
core size 122 of 198  vol = 58.758
core size 121 of 198  vol = 56.354
core size 120 of 198  vol = 54.129
core size 119 of 198  vol = 51.611
core size 118 of 198  vol = 49.502
core size 117 of 198  vol = 48.045
core size 116 of 198  vol = 46.489
core size 115 of 198  vol = 44.599
core size 114 of 198  vol = 43.185
core size 113 of 198  vol = 40.978
core size 112 of 198  vol = 39.029
core size 111 of 198  vol = 36.363
core size 110 of 198  vol = 33.953
core size 109 of 198  vol = 31.343
core size 108 of 198  vol = 29.304
core size 107 of 198  vol = 27.188
core size 106 of 198  vol = 25.655
core size 105 of 198  vol = 24.03
core size 104 of 198  vol = 22.504
core size 103 of 198  vol = 20.906
core size 102 of 198  vol = 19.803
core size 101 of 198  vol = 18.215
core size 100 of 198  vol = 15.652
core size 99 of 198  vol = 14.762
core size 98 of 198  vol = 11.574
core size 97 of 198  vol = 9.391
core size 96 of 198  vol = 7.327
core size 95 of 198  vol = 6.123
core size 94 of 198  vol = 5.592
core size 93 of 198  vol = 4.689
core size 92 of 198  vol = 3.66
```

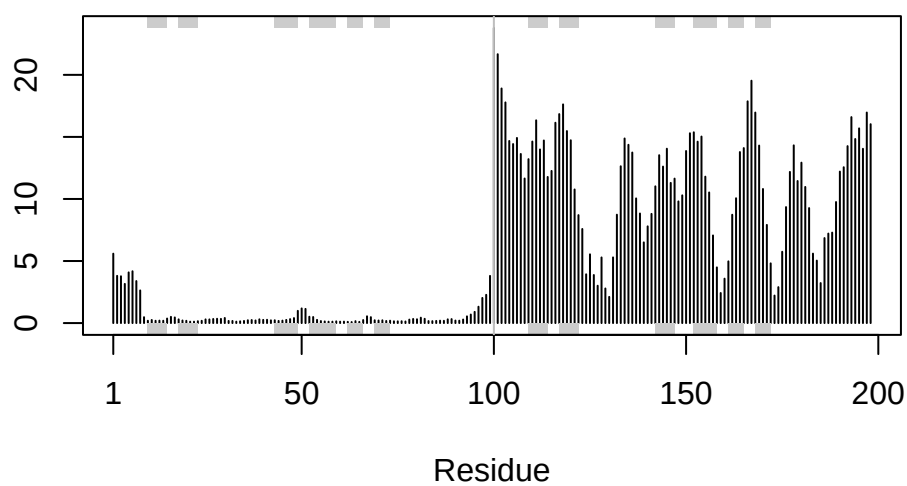
```
core size 91 of 198  vol = 2.779
core size 90 of 198  vol = 2.147
core size 89 of 198  vol = 1.725
core size 88 of 198  vol = 1.158
core size 87 of 198  vol = 0.88
core size 86 of 198  vol = 0.682
core size 85 of 198  vol = 0.527
core size 84 of 198  vol = 0.37
FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

```r
rf <- rmsf(xyz)
plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



##Predicted Alignment Error for domains

```
library(jsonlite)
# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
pattern=".*model.*\\.json",
full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)
attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

$names

```
# Per-residue pLDDT scores
# same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

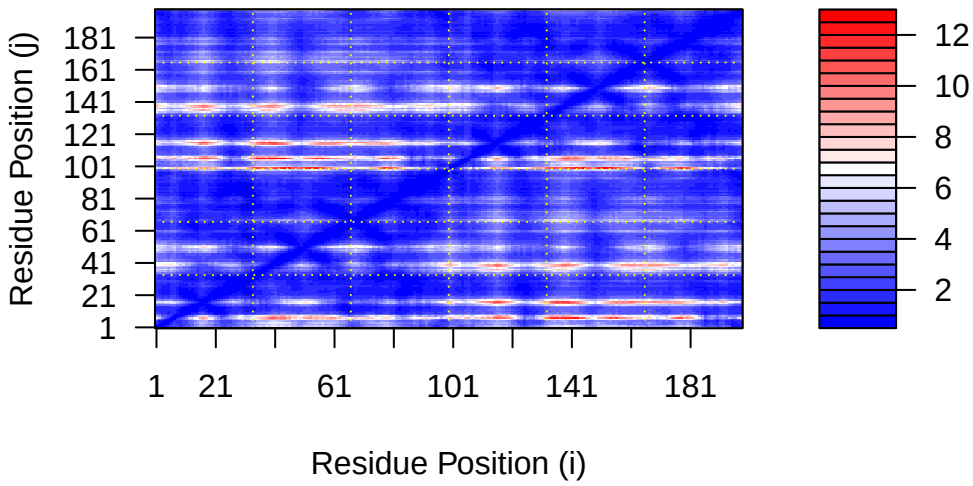
```
pae1$max_pae
```

```
[1] 12.84375
```

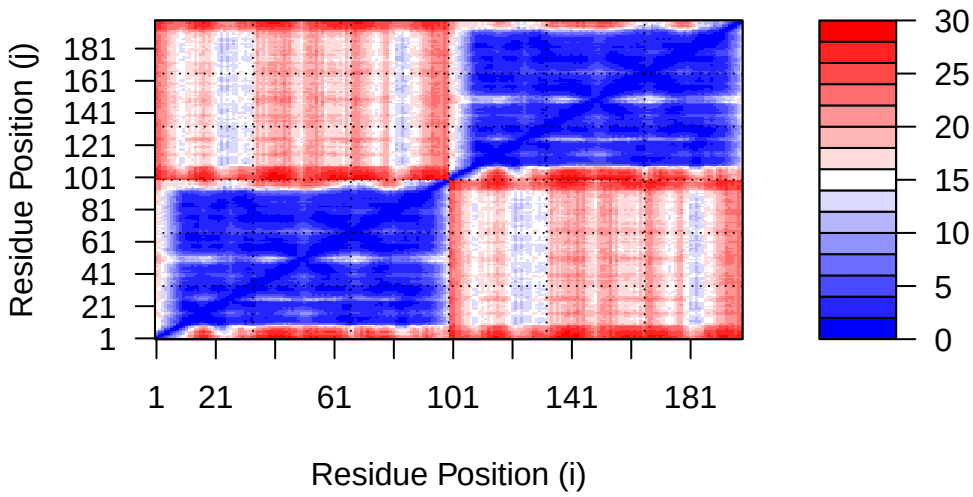
```
pae5$max_pae
```

```
[1] 29.59375
```

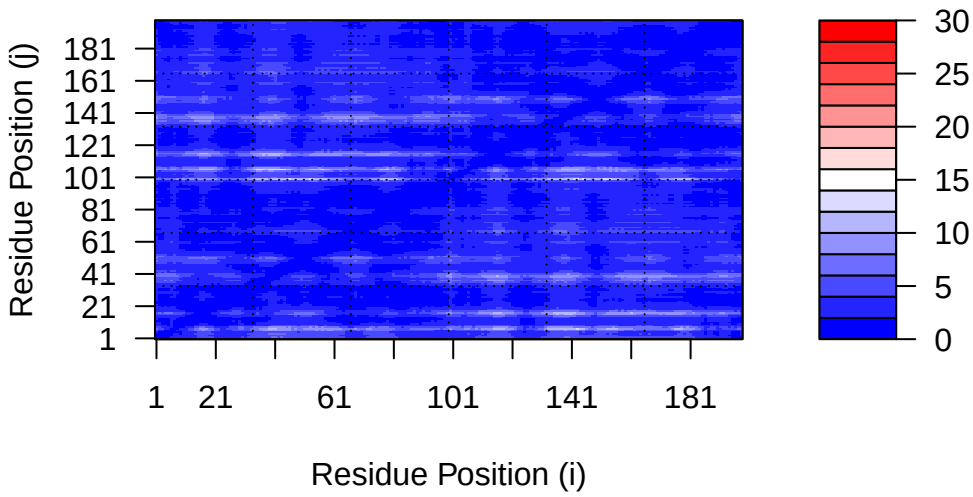
```
plot.dmat(pae1$pae,
xlab="Residue Position (i)",
ylab="Residue Position (j)")
```




```
plot.dmat(pae5$pae,
xlab="Residue Position (i)",
ylab="Residue Position (j)",
grid.col = "black",
zlim=c(0,30))
```




```
plot.dmat(pae1$pae,
xlab="Residue Position (i)",
ylab="Residue Position (j)",
grid.col = "black",
zlim=c(0,30))
```



##Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
pattern=".a3m$",
full.names = TRUE)
aln_file
```

```
[1] "hivpr_23119//hivpr_23119.a3m"
```

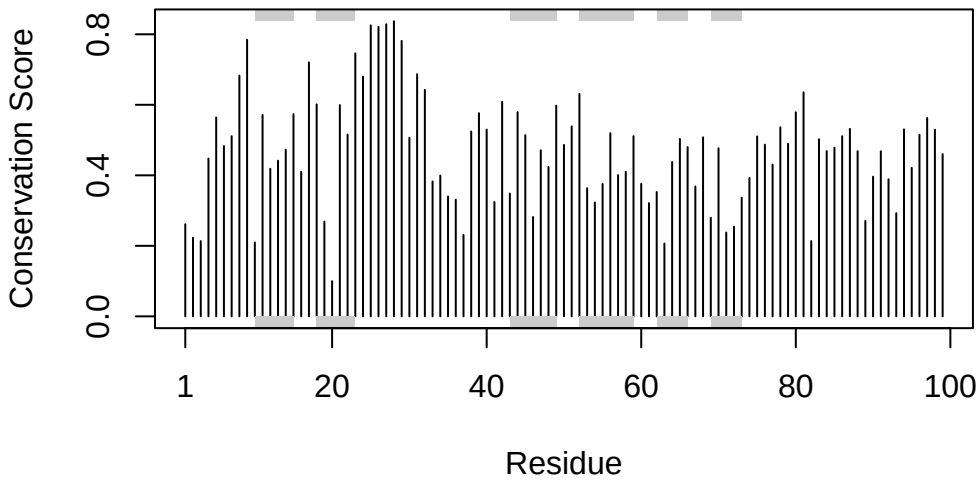
```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397  132
```

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
ylab="Conservation Score")
```




```r
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```r
pdb_files <- list.files("hivpr_23119/", pattern="\\.pdb$", full.names=TRUE)
pdb_files
```

```
[1] "hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
[2] "hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
[3] "hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
[4] "hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
[5] "hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
```

```r
m1.pdb <- read.pdb(pdb_files[1])
```

```r
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

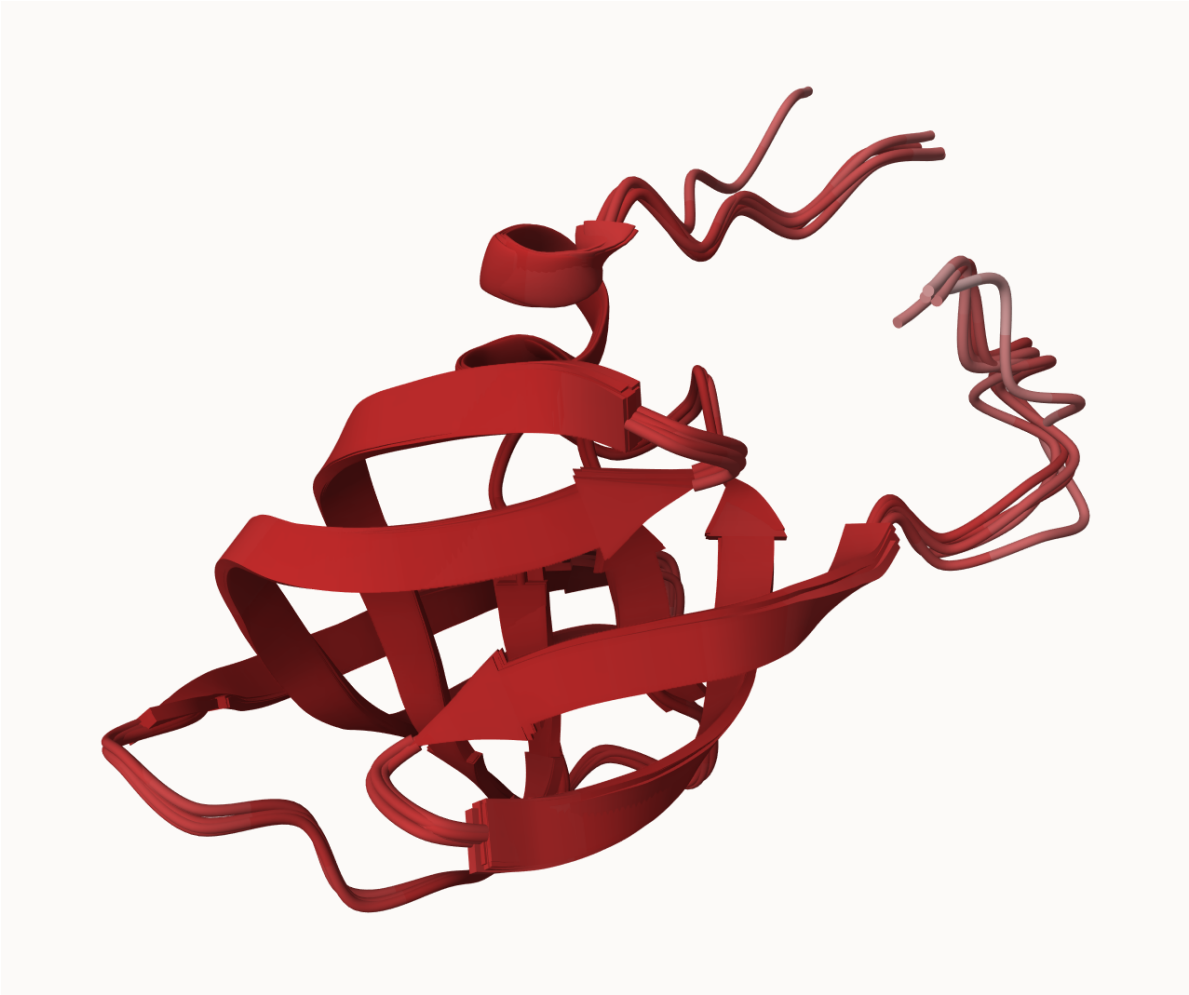


Figure 1: HIVPr Monomer

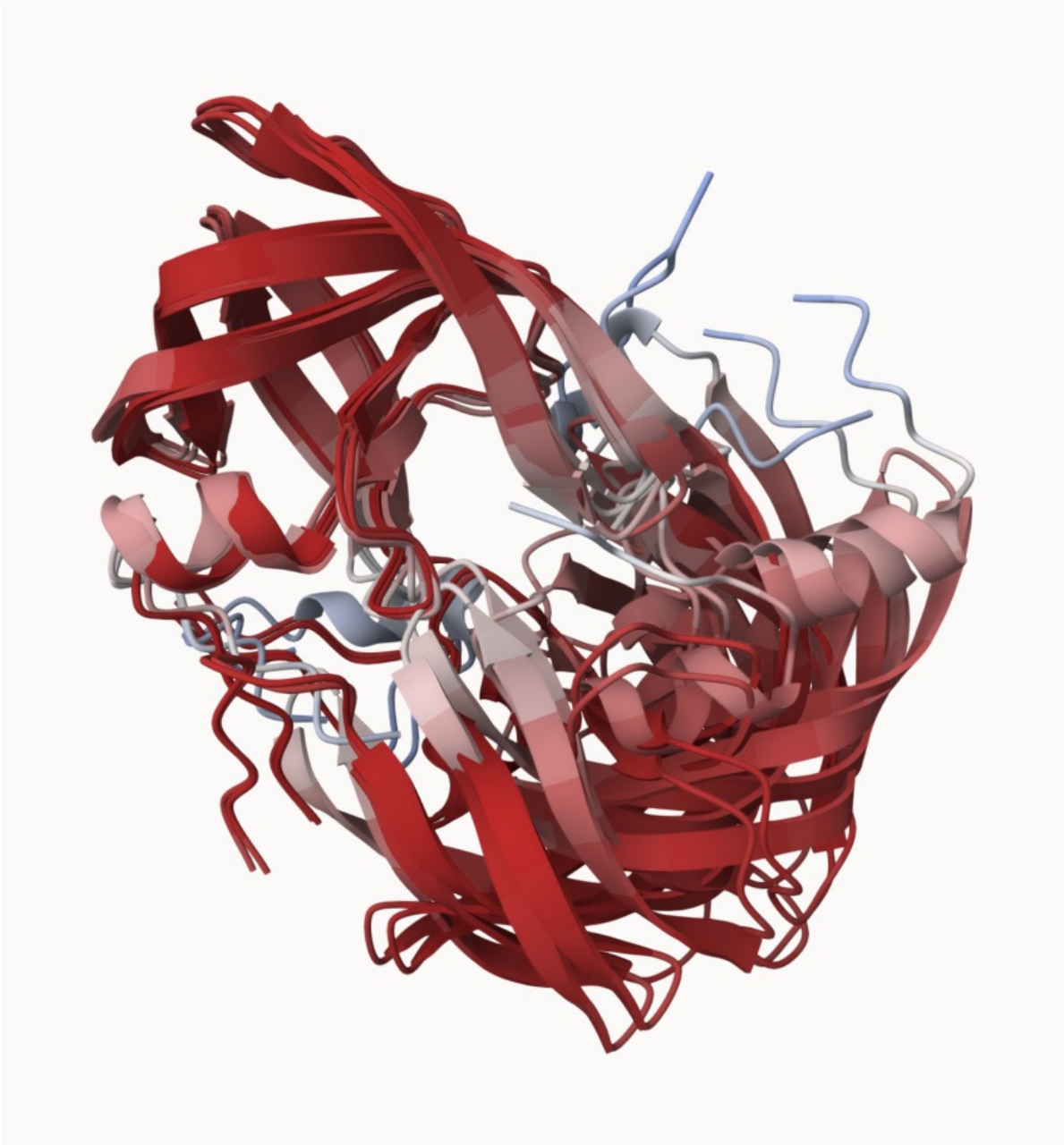


Figure 2: HIV Dimer Aligned to Chain A

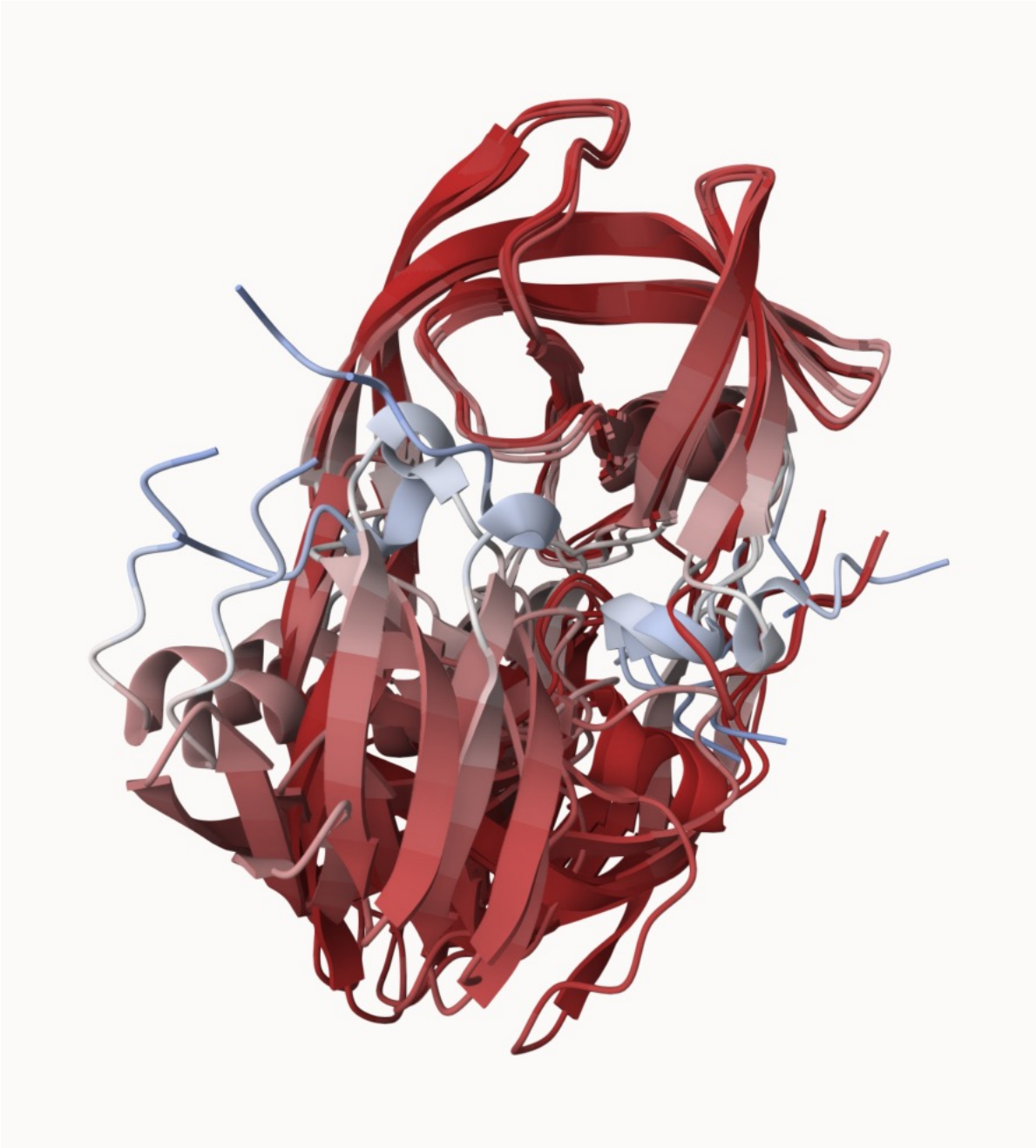


Figure 3: HIV Dimer Aligned to Chain B